

HAB Camp Participant Guide



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Schedule

Week 1

Monday July 27th

- 9:00 AM Camp Arrival & Intro to Camp
- 9:30 AM Physics of Balloon Flight Activity
- 12:00 PM Lunch (Provided)
- 12:45 PM Introduction to Arduino
 - Blinking an LED
 - Reading a Potentiometer
 - Temp Sensor
 - RGB LED
 - Multiple LEDs
 - Temp Sensor
 - Photoresistor
 - Write to SD
- 3:00 PM Initial Project Brainstorm
- 4:00 PM Wrap Up

Tuesday July 28th

- 9:00 AM Camp Arrival, Team selection, and mission description
- 10:00 AM Intro to Data Analysis
- 11:30 AM Payload build time
- 12:00 PM Lunch
- 12:45 PM Payload Build Time
- 4:00 PM Wrap Up

Wednesday July 29th

- 9:00 AM Camp Arrival, Payload Build
- 12:00 PM Payload Build
- 12:45 PM Payload Build
- 2:00 PM Initial Payload testing

4:00 PM Wrap Up

Thursday July 30th

9:00 AM Camp Arrival, Payload Build
11:00 AM Final Payload and analysis tests
12:00 PM Lunch
12:45 PM Flight predictions and launch procedures
2:30 PM Ready payloads for launch
4:00 PM Wrap Up

WEEK 2

Monday August 3rd (launch day)*

7:00 AM Arrive on campus, grab launch equipment
7:15 AM Everyone in vans, start driving
10:00 AM Arrive at CWU, prepare for launch
11:00 AM Launch
12:00 PM Lunch (provided)
1:00 PM Leave CWU to retrieve payload
6:00 PM Dinner (provided)
8:00 PM Check in to hotel (Holiday Inn Express in Ellensburg)
10:00 PM Lights out

Tuesday August 4th (back up launch day)**

8:00 AM Breakfast (provided)
10:00 AM Check out of hotel
10:30 AM CWU campus tour
12:30 PM Lunch (provided)
1:30 PM Depart Ellensburg
4:00 PM Arrive back at Pierce College, Wrap Up

Wednesday August 5th

9:00 AM	Camp Arrival, Data analysis
12:00 PM	Lunch
12:45 PM	Presentation preparation
4:00 PM	Wrap Up

Thursday August 6th

HAB Camp families are invited to join campers for the final presentation ceremony.

9:00 AM	Prepare final presentations
12:00 PM	Lunch (provided by sponsor Ezee fiber)
1:00 PM	Science Dome show
2:00 PM	Final presentations
3:00 PM	Awards Ceremony
3:30 PM	Wrap up

*Launch schedule is tentative and the time we return to campus will depend on overall balloon flight and retrieval time.

**If launch conditions do not look favorable for Monday, launch will take place on Tuesday instead with the same schedule as Monday. Monday's activities will be done on Tuesday instead. Wednesday schedule adjusted as needed.

Physics of Balloon Flight

Why do balloons rise? Balloons rise because of the buoyancy force.

Newton's Second Law: The acceleration of an object is equal to the sum of all forces acting on the object dividing by the mass:

$$\text{Net Force} = \text{mass} \times \text{acceleration}$$

OR

$$F = m \times a$$

Acceleration describes the rate that an object's velocity is changing:

$$\text{acceleration} = \frac{\text{final velocity} - \text{initial velocity}}{\text{final time} - \text{initial time}}$$

$$\text{or } a = \frac{v_{\text{final}} - v_{\text{initial}}}{t_{\text{final}} - t_{\text{initial}}}$$



Example: If you are sitting in a chair, you have a constant velocity of zero. Since your velocity is not changing with time your acceleration is also zero. You have zero acceleration *not* because there are no forces acting on you, but because the forces acting on you are balanced. *The sum of forces acting on you is zero.*

Forces acting on a balloon:

Weight or Force of Gravity: the force of the Earth's gravity pulling down on an object. The strength of the weight force is described by the following formula,

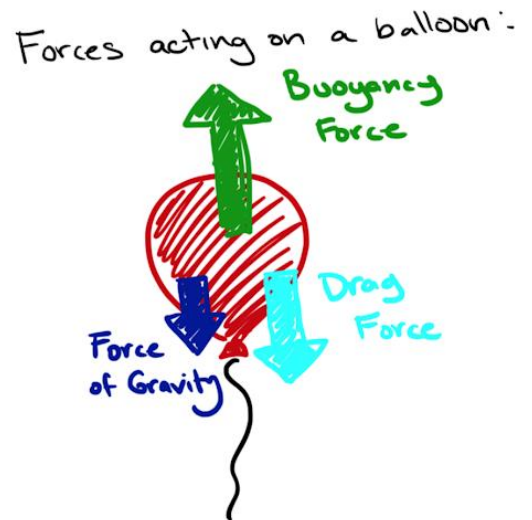
$$\text{Force of gravity} = \text{mass} \times \text{gravitational field}$$

or

$$F_g = m \times g,$$

where mass is measured in units of kg and g is the gravitational field is equal to 9.8 m/s^2 .

Buoyancy Force: Why is there upward force acting on a balloon? That has to do with something called pressure. Pressure is the amount of force acting on a specific area of an object (like force per square centimeter).



$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

The top surface of an object floating in the atmosphere experiences a little less pressure than the bottom surface of the object. Essentially, the air molecules at the bottom of the balloon are punching it with more force per square centimeter than the air molecules at the top of the balloon. This difference causes the balloon to feel a net upward force. The strength of the buoyancy is:

Buoyancy Force = mass density of air $\times$ gravitational field $\times$ volume of balloon

$$\text{Or} \\ F_B = \rho g V$$

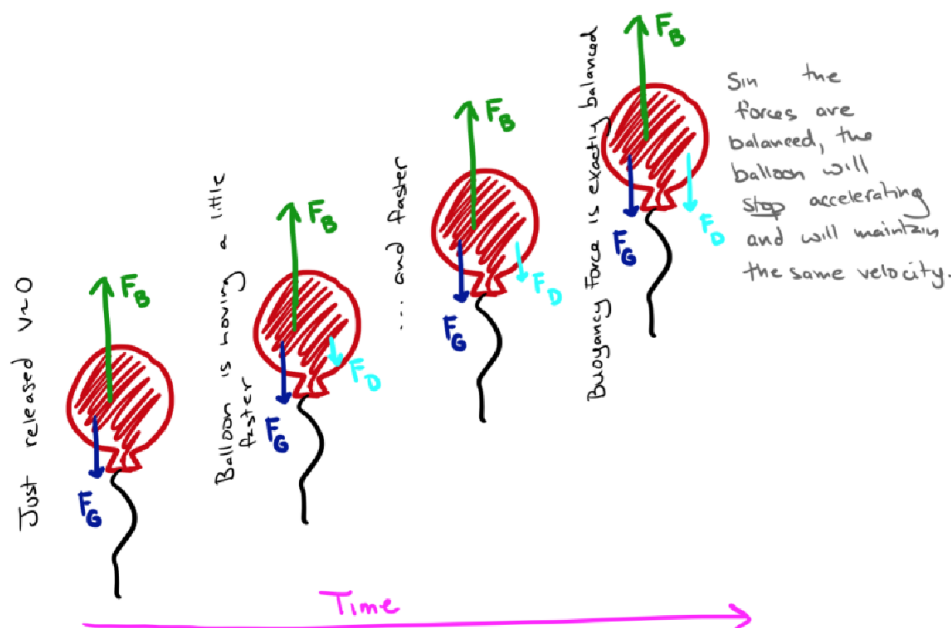
where the Greek character ρ represents the mass density of air (1.225 kg/m³) and V is the volume of the balloon. What does mass density mean? It just means that 1 square meter of air has a mass of 1.225 kg.

Drag Force: A force that acts on any object moving through the air. As the balloon moves through the air it has to push the air molecules out of the way. The strength is given by

Drag Force = some number $\times$ velocity²

$$\text{Or} \\ F_D = C \times v^2,$$

where C is just some number that depends on the balloon's shape and smoothness.



Terminal Velocity:

Eventually, the drag force increases enough that it balances the other forces and at this point the object stops accelerating OR the velocity stops changing. If the velocity stops changing, the drag force also stops increasing and it continues to balance the other forces. At this point the object does not stop moving, it just continues to move at the same velocity, known as the terminal velocity.

Part I: Find the lift of your balloon

The lift force of your balloon is defined as the buoyancy force minus the weight of the balloon, the helium in the balloon, and any attached mass.

$$\text{Lift Force} = \text{Buoyancy Force} - \text{Force of Gravity on balloon}$$

or

$$F_L = F_B - F_G$$

You should have a filled balloon attached to your table. If you add mass to the end of the balloon, you will eventually increase the force of gravity until exactly balances the buoyancy force. This is what is called neutral buoyancy.

Instructions:

1. Add paperclips to the end of the string of your balloon until it is neutrally buoyant.
2. Determine the mass of the paper clips you added using the mass balance.
3. In the space below, use the mass you added to calculate the strength of the *Lift Force*.

Mass of paper clips added (in kg): _____

Strength of the lift force is equal to the force of gravity on the paper clips:

$F_L = \text{mass of paper clips (in kg)} \times 9.8 \text{ m/s}^2 =$ _____

Part II: Modeling the Drag Force

In this part of the lab, you will try to figure out the formula for the drag on your balloon by measuring the terminal velocity of a dropped balloon filled with air.

We know the formula is something like this:

$$\text{Drag Force} = \text{some number} \times \text{velocity}^2$$

Or

$$F_D = C \times v^2$$

where C is just some number that depends on the balloon's size, shape, and smoothness. In this part of the activity, you will try to determine what C is for your balloon.

If we use a balloon with air in it, we can simplify the problem by eliminating the buoyancy force. When a balloon filled with air is dropped, it reaches terminal velocity when the drag force is equal to the force of gravity on the balloon.

$$F_D = F_g$$

Or

$$C \times v_{\text{terminal}}^2 = m \times g$$

If you know how much mass there is and the terminal velocity you can solve for the constant C because g is just a constant (equal to 9.8 m/s^2).

$$C = \frac{m_{\text{tot}} \times g}{v_{\text{terminal}}^2}$$

Instructions:

1. Inflate a balloon until it is approximately the same size as the balloon with helium in it (you can use a string to check the diameter).
2. Drop the air-filled balloon from the 4th floor of the winter garden. Measure the time it takes to fall from the 3rd floor railing to the 2nd floor ground.
3. Take three measurements with the balloon, then add a different amount of mass to the balloon by attaching paper clips. Take three measurements with each mass. Test at least 4 different masses.
4. Use the table below to calculate a value for C for each mass.
5. Calculate the average of all your values of C in the table

Data:

Calculations:

Mass (convert to kg)	Time to fall between 3 floor railing and 2nd floor (in seconds) <i>take 3 measurements for each mass</i>	Average time to fall (s)	Average Terminal Velocity (in m/s) <i>divide distance, 5.11 m, by average time to fall</i>	$C = \frac{mass \times g}{v_{terminal}^2}$ (calculate using average terminal velocity and mass)

Average value of C for all masses: _____

Part III: Balloon Flight Challenge

The goal: Add weight to your balloon so that it rises 9.4 m in 10 seconds. Assume the balloon will reach terminal velocity instantly and will move with that constant velocity after that.

Background:

The lift force of the balloon alone is equal to the value you found in Part I of the lab. If you add extra weight to the end of the string, the force of gravity acting on the additional mass will reduce the lift force. At terminal velocity, the lift force is exactly equal to the drag force:

Drag Force = New Lift force = Original lift Force – force of gravity on additional mass
or

$$C \times v_T^2 = F_{L,original} - m_{additional} \times g$$

You can use this equation to find how much additional mass you need to add to the balloon, so it rises at the correct velocity to travel 9.4 m in 10 seconds.

Steps:

1. Solve for the terminal velocity needed to have the balloon rise 9.4 m in 10 seconds.

Desired terminal velocity (v_T): _____

2. Rearrange the equation above to so that you can use it to determine how much additional mass you need to add to the balloon to have it rise at the correct velocity ($m_{additional}$ should be on the left-hand side of the equal sign). Write the equation below.

$$m_{additional} =$$

3. Now plug in the known values for $F_{L,original}$, C , v_T , and g to calculate a value for $m_{additional}$ in kg.

$$m_{additional} = \text{_____ kg}$$

4. Now use the scale to determine how many paperclips (or other items) you need to add to your balloon.
5. You may not be able to get the necessary mass exactly. What is the mass you actually added?

$$m_{actually\ added} = \text{_____ kg}$$

6. At the end of the session everyone will bring the balloons outside and we will see what team is closest to having the correct amount of weight.

Time for your balloon to rise: _____

OPTIONAL:

Based on the mass you actually added.

1. What is the expected terminal velocity of your balloon?

$$v_T = \text{_____ m/s}$$

2. How much time would it take for the balloon to rise this terminal velocity?

$$t_{\text{rise,new}} = \text{_____ s}$$

3. How close is this to your actual time to rise?

$$t_{\text{rise,actual}} - t_{\text{rise,new}} = \text{_____ s}$$

Introduction to Arduino

This activity will give you an introduction to creating your own devices with small microcontrollers (simple computers) known as Arduino's. Arduino is an open-source platform that allows you to create and control electronic projects, with both hardware and software components designed to make it easy for people to build and program their own interactive devices.



Arduino Board: The Arduino board is a printed circuit board (PCB) that contains a microcontroller, which is a tiny computer that can be programmed to perform specific tasks and other components to make it so the Arduino can be powered correctly and interface with other electrical components.

Programming Language: Arduino uses its own programming language, which is based on C/C++. However, it is simplified and beginner-friendly, making it easier for students to understand and use.

Integrated Development Environment (IDE): To write, compile, and upload your code to the Arduino board, you use the Arduino Integrated Development Environment (IDE). It's a software application that provides a user-friendly interface for writing code, managing projects, and uploading code to the board. The IDE also includes useful tools for debugging and monitoring your projects.

Electronic Components and Sensors: Arduino projects often involve connecting electronic components and sensors to the Arduino board. These components can include LEDs (lights), buttons, buzzers, motors, temperature sensors, motion sensors, and many others. By connecting these components to the appropriate pins on the Arduino board, you can interact with and control them using your code.

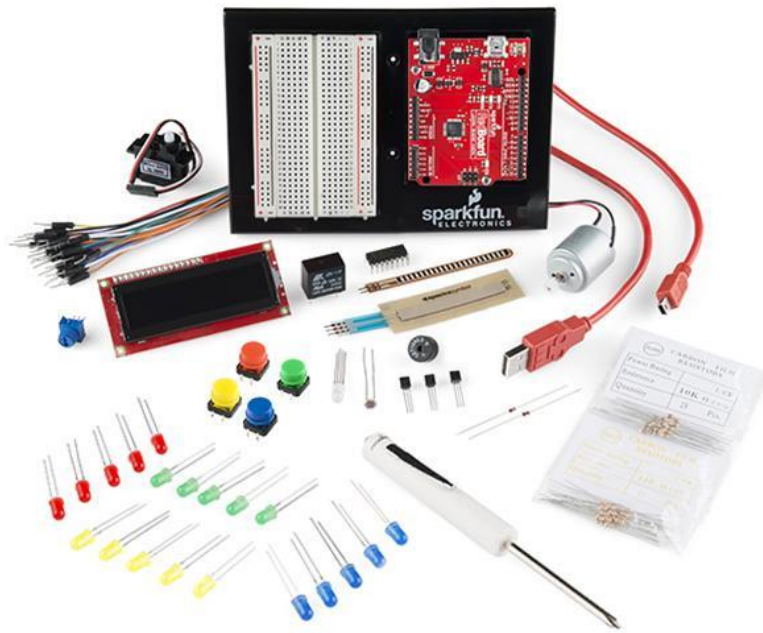
Libraries: Arduino libraries are collections of pre-written code that simplify the process of using specific components or functionalities. Libraries provide ready-to-use functions and commands that can be easily incorporated into your own projects. The sensors or communication devices that are often used with Arduinos with other devices come with libraries that need to be downloaded so that the Arduino knows how to communicate with them.

Examples: The Arduino IDE comes with common example programs that are a useful place to start when trying to build your own project. We will be using example programs in this tutorial.

Getting Hands-on with Arduino

We have selected a collection of Arduino experiments to get you started. You should work with a partner or two. Get as far as you can. Make sure to read the directions carefully. These devices are only as smart as the engineer using them. If a single wire is off, your circuit will not work!

We will be using SparkFun's Inventor's Kit V3.3 to introduce the basic concepts of the Arduino.



To get started, take out the guidebook inside the kit and get a quick understanding of the pinouts of the RedBoard and the anatomy of the breadboard on pages 15-17.

For an online version of the guide, go to:
<https://learn.sparkfun.com/tutorials/sik-experiment-guide-for-arduino---v33>

SIK V3.3	Topic	Page #	Completed?
Experiment 1:	Blinking an LED	19-24	
Experiment 2:	Reading a Potentiometer	25-28	
Experiment 3:	Multiple LEDs	33-36	
Next Day			
Experiment 4:	Choice:		
Experiment 5:	Choice:		
Experiment 6:	Write to SD Card	N/A	

List of Available Sensors

BMP-180

Breakout Board

- Temperature
- Pressure
- Can calculate altitude

DS18B20

Probe

- Temperature

SEN-11345

Standalone Board

- Geiger Counter
- Measures gamma, beta and alpha particles

SEN-12642

Breakout Board

- Microphone/sound detector

ISL29125

Breakout Board

- RGB Values (Visible Spectrum)

GUVA-S12SD

Breakout Board

- UV intensity within a spectrum
- UV Index

MLX90393

Breakout Board

- Magnetic Field Strength

GY271

Breakout Board

- Magnetic Field Strength
- Heading

KODAK PIXPRO SP360 4K

Camera

- Video and Timelapse
- Wide-angle 360° lens
- 4K recording
- Only 2 available

Audio Moth

Standalone Board

- Records high quality audio
- Can record sound beyond what humans can hear

Buzzer

Standalone Part

- Creates a sound at a specific frequency

AS7331

Breakout Board

- Spectral UV Sensor

Mission Objective

What do you want to observe?

What questions do you want to answer?

How are you going to observe it?

What do you expect to happen?:

Team Selection and Mission Description

Team roles:

- *Science/Physics lead*: Takes charge on plotting/analyzing data from payload
- *Mechanical lead*: Takes charge on building structure for airship and HAB payload
- *Code/Electronics lead*: Takes charge on programming and wiring HAB payload
- *Mission director*: Responsible for producing final version of your team's presentation describing the mission, ensuring the team makes good progress in each of the events, and communicating with the higher-ups (camp mentors and directors)

The name of my team is:

My role on the team is:

My teammates and their roles are:

My team's motto is:

Team Mission Patch

List some ideas for ideas, words, and text to put on your mission patch that are

Important to me	Important to my community	Related to balloons, the atmosphere, or near space

Discuss with your team:

- What shape will your mission patch be?
- What will be the primary image?
- What will be in the background?
- What other imagery do we want to include?
- What should you write on your mission patch?

Draw your ideas below:

Idea 1:	Idea 2:
Idea 3:	Idea 4:

You will work with your team throughout the week to come up with your final design!

Present Your Team's Mission!

At the end of camp, you will present what you have learned and show the fruits of your labor. Being able to effectively present your experiments and research is crucial in the world of science and engineering!

Details:

- Each presentation will be 10 minutes long with 5 minutes for questions.
- Every team member should speak.
- Teams will be provided with a presentation template! (in master folder)

What is in the presentation?

- An introduction of the team members.
- An outline of what you learned during the week and what you will present.
- How your team did in the physics of balloon challenges.
- A picture of the patch your team designed.
- A picture of your payload box and a description of what you measured.
- A description (graph) of any data collected during the flight.
- Your favorite images or videos from flight or camp.
- Anything else you want, as time allows.

Speaking notes for the presentation:

HAB Payload Build

Now that you've set your objectives, it's finally time to build the payload! Your goal is to design a payload that is both lightweight and sturdy to withstand harsh turbulence throughout its flight.

FAA regulations:

FAA rules apply to balloons that meet the descriptions below. We will stay under these limits so that the additional restrictions do not apply.

For any unmanned free balloon that-

- Carries a payload package that weighs more than four pounds and has a weight/size ratio of more than three ounces per square inch on any surface of the package, determined by dividing the total weight in ounces of the payload package by the area in square inches of its smallest surface;
- Carries a payload package that weighs more than six pounds;
- Carries a payload, of two or more packages, that weighs more than 12 pounds; or
- Uses a rope or other device for suspension of the payload that requires an impact force of more than 50 pounds to separate the suspended payload from the balloon.

In our case, each payload must be under 2.5 pounds.

Your payload should include:

Your Arduino Board and Sensors required for measurements. A battery pack is required to power your Arduino.

Secured String to attach a parachute or to chain under another payload.

- When attaching a parachute, connect strings to the guide ring and the parachute on the top of the guide ring.
- When chaining together, tie one end to a hook attached to your payload and have the other end attach to a hook on another payload
- Each payload should have straps with hooks.

Optional:

Camera + extra battery pack. If you want to record a video of the flight of

your payload, take **only one** of the two cameras we have. You will also need an extra battery pack to power the camera (unless you want your footage to end prematurely...)

Other fun stuff: Any light and non-hazardous items that you want to send up so you can show it off to your friends! However, make sure it's something you're okay with losing. It's possible your payload may become unretrievable.

Consider: Operating Conditions

Your delicate electronics will be subject to harsh environments. Consult each part's technical specifications for details such as operating temperature to make sure your electronics can reliably operate during its entire flight.

If your part does not need to be exposed to the environment to function, consider keeping it insulated inside your payload.

If your part does require being exposed to the environment but may be exposed to conditions outside its operating conditions, expect inaccurate data. Once returned to its normal operating condition, the part should function as normal.

High Altitude Balloon Parachutes

2ft High Altitude Balloon Parachute... Price: \$35.50 / Colors may vary

Weight of parachute: 1.5oz

Packing Volume: $2.6''D \times 1.5'' = 7.96''^3$

Descent Rate:

15ft/sec: .70lbs

17ft/sec: .90lbs

20ft/sec: 1.13lbs

25ft/sec: 1.5lbs

5ft High Altitude Balloon Parachute... Price: \$55.50 / Colors may vary

Weight of parachute: 5.0oz

Packing Volume: $3.0''D \times 4.0'' = 28.27''^3$

Descent Rate:

15ft/sec: 4.7lbs

17ft/sec: 5.7lbs

20ft/sec: 8.0lbs

25ft/sec: 12lbs

3ft High Altitude Balloon Parachute... Price: \$45.50 / Colors may vary

Weight of parachute: 2.4oz

Packing Volume: $2.6''D \times 2.25'' = 11.94''^3$

Descent Rate:

15ft/sec: 1.7lbs

17ft/sec: 2.1lbs

20ft/sec: 2.8lbs

25ft/sec: 4.4lbs

6ft High Altitude Balloon Parachute... Price: \$60.50 / Colors may vary

Weight of parachute: 6.0oz

Packing Volume: $3.0'' \times 5.0'' = 35.34''^3$

Descent Rate:

15ft/sec: 6.5lbs

17ft/sec: 8.0lbs

20ft/sec: 11.5lbs

25ft/sec: 17.0lbs

4ft High Altitude Balloon Parachute... Price: \$50.50 / Colors may vary

Weight of parachute: 3.5oz

Packing Volume: $2.6''D \times 3.5'' = 18.58''^3$

Descent Rate:

15ft/sec: 3.0lbs

17ft/sec: 3.7lbs

20ft/sec: 5.0lbs

25ft/sec: 7.6lbs

7ft High Altitude Balloon Parachute... Price: \$70.50 / Colors may vary

Weight of parachute: 8.0oz

Packing Volume: $3.0'' \times 6.5'' = 45.94''^3$

Descent Rate:

15ft/sec: 9.0lbs

17ft/sec: 11lbs

20ft/sec: 15lbs

25ft/sec: 23lbs

Create a Timeline

It is important to stay on track during the limited time we have to plan, build, and test your payload. Use the table below to set goals on what you plan to have done by each day.

Day	Goal
Tuesday	
Wednesday	

Thursday	
----------	--

Draw a Schematic

Drawing a rough sketch on how your payload will look helps teammates work with a uniform vision on how the final product should look like. Make sure to include all parts and label them. It's best to have multiple angles as well!

Challenge (Optional)

It is very likely that your payload will sway violently during flight (though not detrimental). Design active/passive measures to keep your payload as stable as possible.

Testing

Now that you have your first prototype, it's time to confirm it will survive up in the air.

Construction

To make sure nothing will tear-off and ruin everything, it is important that everything is secured. Try...

- Shaking your payload to check for anything loose
- Tugging the connecting string to make sure it won't cut through the payload

Electronics

It would be horrible to find that your Arduino was running well for the first ten minutes, but then suddenly bugged out for the rest of the flight. Conduct a longevity test for your electronics to make sure they'll last for the entire flight. Check for...

- Connectivity: Make sure a small jostle won't wiggle a connection loose.
- SD data safety: Test for a scenario where the Arduino/SD card loses power. Check if the data will still be intact and not corrupted.
- Longevity: Test if your hardware will run for at least more than 10 minutes.

Vacuum Chamber

You will have access to a vacuum chamber to simulate the change in pressure.

IF YOU HAVE A CAMERA

Make sure you are using the video settings you want. Understand filming orientation and recording mode.

Refer to the Owner's Manual on your flash drive for documentation on each setting.

File name: sp360_4k-manual-en.pdf

Test	Describe the test	Results/ What needs improvement?
Construction /Security		
Code		
Data Security		
Battery Longevity		
Temperature		
Pressure		

Customize Your Payload!

Stickers, drawings, signatures you name it! Now that your payload is all tested and ready to go, add some flair to your boring little box!

You should add a contact number in case someone finds it before you!

MAKE SURE TO CHARGE ALL OF YOUR BATTERIES
OVERNIGHT! LET'S GO!

Flight Predictions

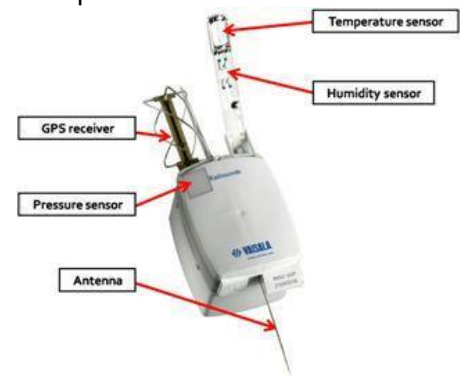
Introduction

Predicting daily global weather requires both careful modeling and data. High altitude balloons are one way that scientists collect data for forecasting and monitoring the atmosphere. In fact, around 1000 balloons are released twice a day at fixed locations worldwide. These balloons, known as radiosondes, transmit the data they collect back to the ground using radio waves. You might have come across the remains of a radiosonde while hiking in the Cascades.

Most high altitude balloon (HAB) payloads do not transmit their data back to the ground immediately. Instead, the data is stored on board and can only be collected if and when the balloon is found. This requirement of retrieving the balloon makes it crucial to carefully consider when and where to launch a high altitude balloon.



Before launching a balloon, you need to use software to predict where it will land. High altitude balloons ascend to great heights in the atmosphere and can enter the jet stream. In Washington, this means they can travel different distances and directions depending on the behavior of the jet stream on that particular day. Fortunately, there are models that can predict atmospheric winds days in advance. Balloon prediction software utilizes these models to understand how the winds will affect the balloon's flight.



Winds are just one aspect of the balloon prediction puzzle. It's also important to determine how high and how fast the balloon will ascend. Balloons rise because of the buoyancy force, and the rate of ascent depends on the strength of that force relative to the weight of the balloon and the drag force. Generally, the more helium you put in a balloon, the more air it displaces, leading to a greater buoyancy force.

As the balloon ascends higher into the atmosphere, it expands due to the decrease in atmospheric pressure around it. The maximum altitude a balloon can reach is determined by the stretching capacity of the material it is made of. A larger balloon, capable of stretching further without breaking, can reach higher altitudes.

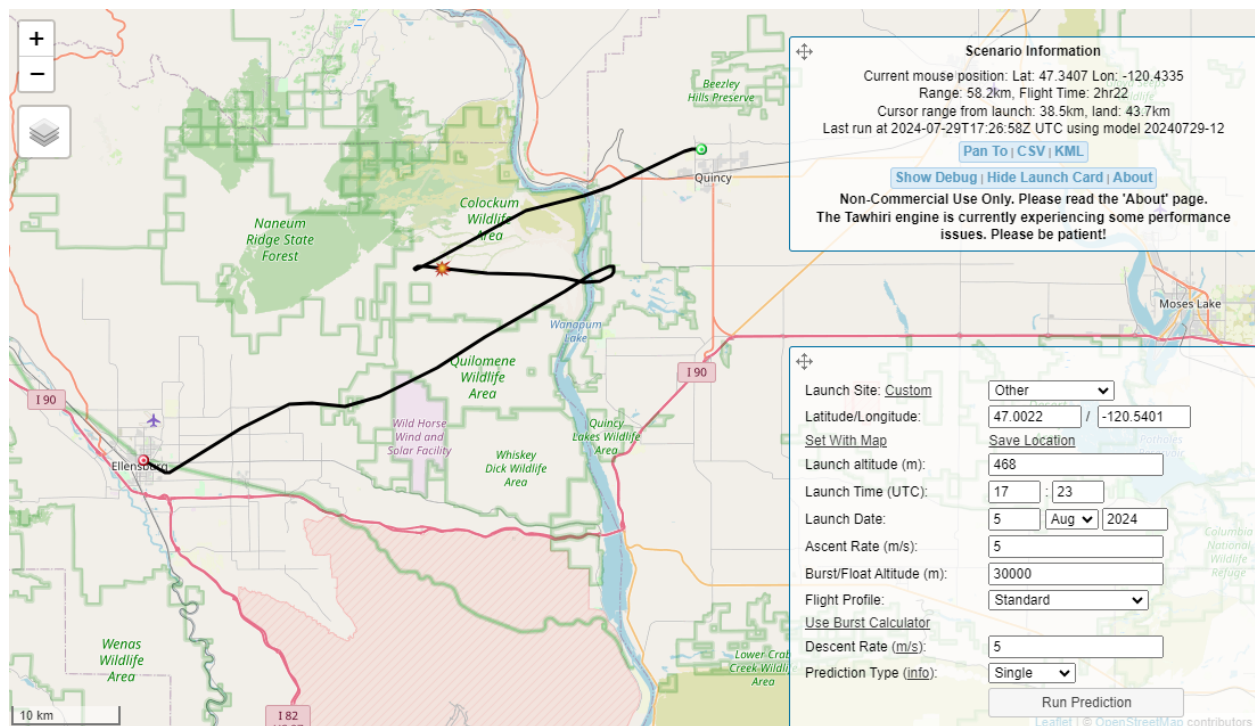
HAB prediction software

There are several websites that use models to help you predict the flight of a high altitude balloon. A good one is the CUSF landing predictor, which you can find at <https://predict.sondehub.org/>

Here's how you can use it:

1. Enter your launch coordinates or select them on the map.
2. Choose the altitude you plan to reach.
3. Select the date and time of your launch. You can use the Google search bar to convert between PST and UTC time.
4. Descent rate will depend on your payload mass and the parachute size. <https://the-rocketman.com/recovery-html/>
5. Click on "run prediction."

Give it a try! Enter the current date and time to see where the balloon will go. You can even draw your prediction on the map. The "Scenario Information" window will show you the total range and duration of the flight.



Using CUSF to determine the lift force

The balloon's trajectory also depends on other factors like the type of balloon, the payload mass, and the amount of helium you use. CUSF can

take all of these parameters into account. If you select the "**Use Burst Calculator**" option in the bottom right window, a new screen will appear.

This window allows you to choose a desired altitude or ascent rate for a specific balloon type (usually around 5 m/s is ideal for ascent rate) and payload mass (not including the balloon). Once you select these parameters, the model runs and provides you with estimates for the actual ascent rate, burst altitude, launch volume, and neck lift. The neck lift represents the amount of mass you would add to the neck of the inflated balloon to make it neutrally buoyant when you have added the correct amount of helium. For a properly inflated balloon, you want it to be able to lift this much mass.

How do you use the neck lift? We usually add roughly that much mass to a bag and attach it to the neck of the balloon as it inflates. Periodically, we check if the balloon can lift the mass on its own. When it can, we know it's ready to go. We then remove the bag of "neck lift" mass, attach the payload (which weighs less). Since the payload weighs less, the balloon will no longer be neutrally buoyant but will ascend at the correct rate. This approach is easier than precisely measuring the volume of the balloon or the amount of helium we have used.

Find the total payload mass, by weighing each payload and recording below.

Payload description	Mass in grams
Parachute & connections	
Total mass:	

Use the Burst Calculator with these parameters to aid your prediction:

Payload Mass (g) : Total mass above

Balloon Mass(g): Will use either 1200g or 1500g balloons

Target Burst Altitude (m): Leave this blank

Target Ascent Rate (m/s): 5

Record the output values below.

Burst Altitude (m): _____

Ascent Rate (m/s): _____

Time to Burst (s): _____

Neck Lift (g): _____

HAB Tracking Methods

Tracking a Balloon with a GPS tracker and Satellites

We will be using a GPS Spot tracker to track our balloon. Spot trackers utilize a GPS receiver to determine the device's precise location by receiving signals from multiple satellites in orbit around the Earth. Once the spot tracker knows its location, it uses another set of satellites to transmit that information to a network of ground stations. They are commonly used for outdoor activities, such as hiking, camping, or expeditions, where reliable communication and location tracking are essential.

It's important to note that Spot trackers rely on satellite connectivity, so they require a clear view of the sky to establish a connection with satellites. This means they need to be facing upwards on your balloon payloads. *Most* commercially available spot trackers are designed to stop working about 60,000 ft or 18.5 km. However, you can buy ones that work at higher altitudes. However, unlike APRS (see below), Spot trackers usually send the balloon's location even after it has hit the ground (as long as it is facing upwards). That is why combining the two types of tracking systems is beneficial.

By carefully monitoring the balloon's location and altitude, you can determine if its actual flight path aligns with the predicted path. Sometimes the prediction is incorrect, or there may have been errors during the inflation process or in measuring the mass of the payload. If you believe the prediction is significantly off, you can use the data to run a new prediction, perhaps with a more accurate ascent rate. On the other hand, if the balloon's predicted path closely matches its actual path, it's time to start heading towards the expected landing location.

Our Spot trackers will report their locations here:

<https://maps.findmespot.com/s/5YXM>

Before launch check that the Spots are connected to the satellites and transmitting data.

Tracking a Balloon with APRS (alternate)

Another way to track a balloon is with Automatic Packet Reporting System (APRS). APRS is a digital communication system used by amateur radio operators to track and exchange information about the location of objects, such as vehicles, weather stations, or even high altitude balloons.

To use this APRS, you need to attach a GPS chip and a radio transmitter to your payload. You can make these transmitters yourself or buy them.

StratoTrack is an example

(<https://www.hightitudescience.com/products/stratotrack-aprs-transmitter>). Additionally, you must have a call sign, which means you or someone you know has passed the amateur radio licenses (HAM) exam.

The APRS system determines and broadcasts your location using GPS satellites and a network of radio towers. Your location is eventually displayed on a server, allowing you to view it on a website. APRS antennas need a clear line of sight to radio towers to broadcast your data. When using them to track balloons they often stop transmitting location information. You can track a balloon with the aprs.fi website and the call sign for the balloon.

Balloon Recovery

Balloon recovery can be tricky, even if the balloon follows its predicted path. Oftentimes recovering the balloon means driving on challenging roads or hiking long distances. It is always good to be patient and careful. You can always come back for the balloon another day.

You should also always ask private landowners for permission before entering their property, even if it is a large farm or ranch. You may ruin crops or walk through toxic pesticides to get to your payload. A landowner or local may also be willing to help retrieve or locate your payload. In case you can't find it right away.



Finally, you should label your payload with your phone number and an address in case you are not able to find it or retrieve it right away.

Balloon Recovery Challenge

Use the data from the flight predictor page to try to predict the landing location of the payload. When you are done, write down the precise

latitude and longitude below. Latitude: _____ Longitude

Launch Procedures

Pre-Launch

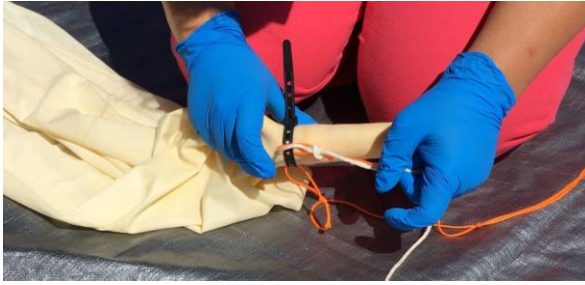
1. Make sure each box has contact info on it. *File NOTAM if necessary.*
2. Check Spot GPS are activated and reporting positions.
3. Seal payloads and ready them for launch. (What does this mean for YOUR payload?)
4. Each team records their payload mass.
5. Calculate total payload mass.
6. Run flight predictions
 - a. Record everyone's prediction, who will be the closest?
 - b. Record needed Neck Lift : _____
7. Place weight equal to the Neck Lift in a heavy duty plastic bag. Attach rope with carabiner to the bag so it can be attached to the balloon firmly, but released easily.

Launch

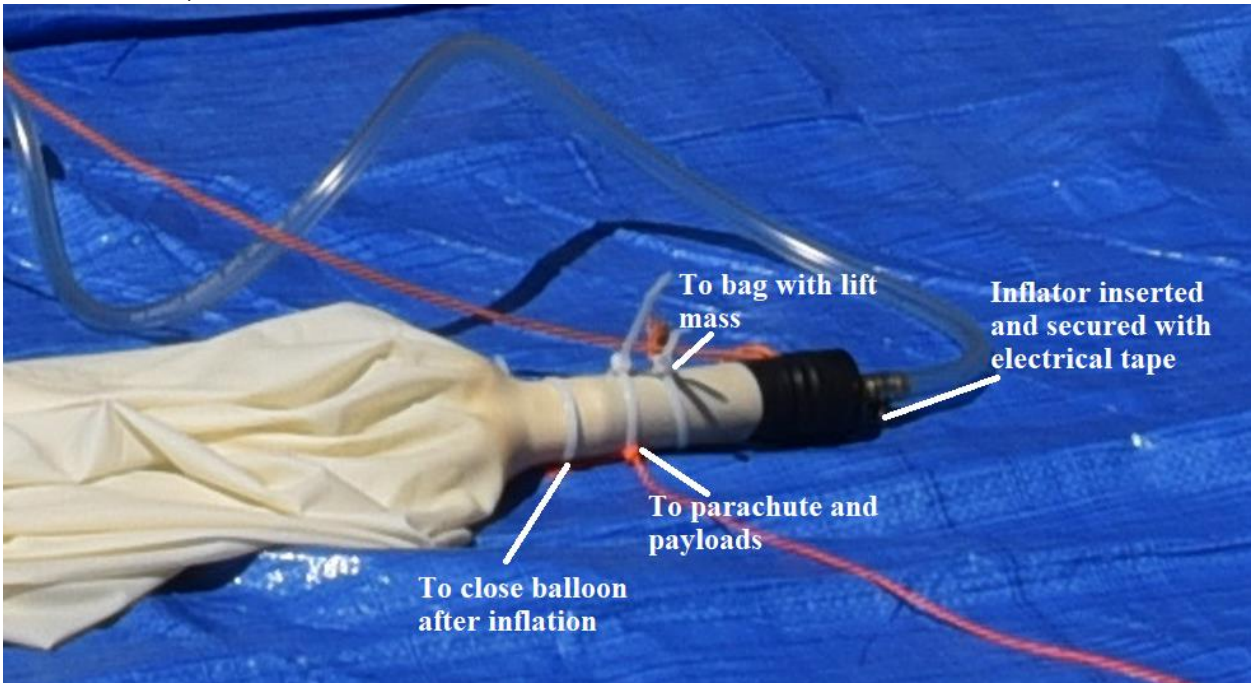
1. Lay tarp on the ground and weigh down the corners so it doesn't fly away. Keep the tarp clean by not wearing shoes on the tarp.
2. Ready the helium.
 - a. Place bottles for filling and secure in place.
 - b. Attach regulator
 - c. Attach balloon inflator
3. All balloon filling assistants put on gloves. Do not touch the balloon with your skin, oils will transfer from your skin and weaken the balloon.
4. Chain payloads together on tarp. Lay out neatly in a line. Make sure the parachute is on top, just under the balloon.



5. Carefully unfold the balloon and lay on the tarp. Secure three heavy duty zip ties around the neck of the balloon. Make sure they are loose enough that the balloon inflator will fit in the neck of the balloon. Tie the payloads to one zip tie, and the neck lift bag to another. The third will be used to seal the balloon after inflation.



6. Insert the balloon inflator into the neck of the balloon and wrap with electrical tape.



7. Carefully pick up the balloon and cradle in the arms of two people. Begin filling the balloon with helium.



8. As the balloon fills, one person needs to hold on to the neck of the balloon firmly so that it doesn't fly away. The filling assistants should surround the balloon and gently hold it in place with gloved hands so it is not whipped around by wind.
9. Occasionally check to see if the balloon can lift up the mass in the bag. When the balloon with the bag attached is neutrally buoyant, you have enough helium!
10. While holding tightly to the balloon:
 - a. Remove the inflator
 - b. Cinch the zip ties. Cut the zip tie ends and wrap with electrical tape.
 - c. Fold over the end of the balloon. Cinch closed with a fourth zip tie, cut end and wrap with electrical tape.



11. Hold on very tightly to the balloon. Remove the bag with neck lift mass.
12. Slowly release the balloon, guiding up the payloads and avoiding any obstacles.

Post-Launch

1. Record launch time: _____
2. Rerun launch prediction
3. Check GPS report <https://maps.findmespot.com/s/5YXM>
4. Wait for the payloads to start descending (hang out, eat lunch, etc.).
Retrieve payloads!

Notifying the FAA

No less than 7 days prior to launch, fill out a [HIBAL Worksheet](#) and email it to the PART 101 coordinator for our region, Fred Johnson, at 9-ato-wsa-osg-part101@faa.gov

File NOTAM (see below) **24-72 hours before launch**.

Make a **30 minutes prior to launch** notification call to Seattle ARTCC (ZSE) Operations Manager at **253-351-3520** and comply with any requests that they may make (i.e. let them know departing 24,000 Feet MSL on Ascent, Departing 24,000 Feet MSL on Descent or payload on the ground notification), if any.

Notice to Airmen (NOTAM)

Although we will be staying under the limits where FAA regulations apply. It is still a good idea to file a Notice to Airmen (NOTAM) if the balloon will take off or land near an airport.

NOTAM Template

Phone number for NOTAM: 877-487-6867

When calling say "I would like to file a NOTAM for an unmanned free balloon."

Flight contact: Hillary Stephens
Number: 253-254-9161

Launch Date:

Launch coordinates: 47.0022, -120.5402 (CWU campus - Discovery Hall)

Launch altitude: 482.3 m (CWU campus)

Launch time*: 11 AM - 12 PM PST, 18:00 - 19:00 UTC

*to convert from PST to UTC, add seven hours

Airport closest to launch site: Bowers Field (ELN: Echo-Lima-November)
(skyvector, <https://skyvector.com/>)

Closest VOR/DME (ELN)

Fix, Radial, Distance to launch site: ELN 227° Radial 3.61 NM Fix (ELN 227004)

Distance and direction of closest airport: 2.01 nautical miles South (google maps)

Landing Location: [Latitude] [Longitude] (description, e.g. East of Moses Lake, East of ellensburg)

Range: [how many km or miles]

Flight Time: 2 hr 30 min

Approx. Landing Time: (e.g. 1pm to 2:30 pm PST, 20:00 - 21:30 UTC)

Max altitude: 100,000 ft

Rate of climb: 5 m/s or 16 ft/s

Airport closest to landing site: [name and three letter sign] (sky vector, <https://skyvector.com/>)

Distance and director of closest airport: [distance and N/S/E/W direction]
(google maps, measure distance)

Flight Prediction Map: [copy and paste]

After providing information they will read it back to you. If correct, say "read back correct." If it isn't correct, fix errors.

Enter this information after calling in the NOTAM

NOTAM Officials Initials and reference number:

Also call Bower's Field Tower ~10 minutes before flight (Midstate): 509-962-7852
(Supposed to call someone else but I don't remember)

Alert CWU Campus Police of launch time and location: (509) 963-2959, 509-925-4008

Phonetic Alphabet:

CHARACTER	MORSE CODE	TELEPHONY	PHONIC (PRONUNCIATION)
A	• —	Alfa	(AL-FAH)
B	— • • •	Bravo	(BRAH-VOH)
C	— • — •	Charlie	(CHAR-LEE) or (SHAR-LEE)
D	— • •	Delta	(DELL-TAH)
E	•	Echo	(ECK-OH)
F	• • — •	Foxtrot	(FOKS-TROT)
G	— — •	Golf	(GOLF)
H	• • • •	Hotel	(HOH-TEL)
I	• •	India	(IN-DEE-AH)
J	• — — —	Juliett	(JEW-LEE-ETT)
K	— • —	Kilo	(KEY-LOH)
L	• — • •	Lima	(LEE-MAH)
M	— —	Mike	(MIKE)
N	— •	November	(NO-VEH-BER)
O	— — —	Oscar	(OSS-CAH)
P	• — — •	Papa	(PAH-PAH)
Q	— — • —	Quebec	(KEH-BECK)
R	• — •	Romeo	(ROW-ME-OH)
S	• • •	Sierra	(SEE-AIR-RAH)
T	—	Tango	(TANG-GO)
U	• • —	Uniform	(YOU-NEE-FORM) or (OO-NEE-FORM)
V	• • • —	Victor	(VIK-TAH)
W	• — —	Whiskey	(WISS-KEY)
X	— • • —	Xray	(ECKS-RAY)
Y	— • — —	Yankee	(YANG-KEY)
Z	— — • •	Zulu	(ZOO-LOO)
1	• — — — —	One	(WUN)
2	• • — — —	Two	(TOO)
3	• • • — —	Three	(TREE)
4	• • • • —	Four	(FOW-ER)
5	• • • • •	Five	(FIFE)
6	— • • • •	Six	(SIX)
7	— — • • •	Seven	(SEV-EN)
8	— — — • •	Eight	(AIT)
9	— — — — •	Nine	(NIN-ER)
0	— — — — —	Zero	(ZEE-RO)